

Towards the assessment of business process knowledge intensity – a systematic literature review

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Abstract

Purpose – The aim of the article is to propose an integrated definition of knowledge-intensive business processes (kiBPs) and a model of business process (BP) knowledge intensity and develop holistic criteria and measures for the assessment of BP knowledge intensity.

Design/methodology/approach – A qualitative systematic literature review (SLR) was carried out as a means of evaluating and interpreting all available research relevant to the research topic, in this case, BP knowledge intensity. The SLR is based on the resources of the ProQuest, Springer Nature and ScienceDirect full-text databases. Regarding inclusion criteria, the authors considered peer-reviewed journals and conference proceedings published in English in the last 10 years. The results were summarized and synthesized narratively.

Findings – Based on the review, the authors classified existing knowledge intensity definitions, starting with knowledge-intensive economy or country, sectors and services, through knowledge-intensive organizations, BPs, to tasks. Then, an integrated definition of kiBPs was proposed and the business process knowledge intensity model encompassing three domains developed. Finally, 12 knowledge intensity criteria with measures were proposed.

Originality/value – The major value of this research is that it presents a novel approach to understanding BP knowledge intensity and kiBPs. By creating the BP knowledge intensity model, supplemented by the assessment criteria and measures, the authors contribute to the more effective management of kiBP, help find similarities and differences between various knowledge-intensive processes, as well as provide important implications for their effective management and research.

Keywords BPM, Business process management, Knowledge management, Business process knowledge intensity, Knowledge-intensive business processes (kiBPs), Systematic literature review

Paper type Literature review

1. Introduction

The knowledge economy (KE) is an economy that is reliant on the production, transfer and utilization of information and knowledge (OECD, 1996). Contrary to mass production-driven economies, in the KE, the competitiveness of most industries is much more dependent on the value of intangible resources and know-how rather than on the physical base of production or raw materials. Thus, in the KE the management of an organization should lead to the most effective acquisition, storage, distribution and, first and foremost, the creation and use of knowledge, with the aim of creating value and building competitive advantage (Davenport and Prusak, 1998; Davenport, 2015; Wu and Hu, 2018). Consequently, organizations are becoming increasingly conscious of the necessity to formalize and capitalize on the knowledge produced and used by their business processes (BPs) in order to improve their competitive advantage (Ben Hassen *et al.*, 2016), as the more knowledge intensive the resources, the less imitable they are and the greater the opportunities for development they generate (Autio *et al.*, 2000).



As each product and service delivered to the market is the result of the execution of BPs, it requires the close relationship between knowledge management and business process management (BPM) (Gronau *et al.*, 2005; Little and Deokar, 2016; Szelągowski, 2019). It cannot be limited to the use of knowledge in the design phase of structured, repeatable BP, as in the case of traditional BPM (Kemsley, 2011; van der Aalst, 2013). In the KE, it must also include processes that require the use in the process execution phase knowledge and the dynamism of knowledge workers in accordance with the requirements of a specific implementation context, e.g. meeting individual customer requirements (Reichert and Weber, 2012). Only then will it be possible to effectively use techniques such as process mining, machine learning, artificial intelligence that allow for the disclosure and codification and further dissemination and use of knowledge created during the implementation of processes (Djurica *et al.*, 2020; van der Aalst, 2016). Processes that are characterized by the need to transform and create knowledge in the whole lifecycle are called knowledge-intensive business processes (kiBPs).

For many years, knowledge-intensive sectors and organizations have enjoyed growing interest thanks to their product and service innovation and their impact on productivity, international competitiveness, the creation of well-paid jobs and overall economic growth. This growing interest reflects the level of employment in knowledge-intensive activities in the EU, which currently exceeds 36%. In countries such as Belgium, Ireland and Switzerland, this indicator exceeds 44% (Eurostat, 2021). However, it is recognized that although the classic BPM approach is suitable for traditional (predefined and standardized) processes, it is not suitable for BPs, the planning and implementation of which are largely dependent on making decisions based on expert knowledge (Dumas *et al.*, 2018). To be part of the digital transformation, companies must combine advanced automation, cloud computing, computer-assisted processes and intelligent algorithms to transform their businesses and stay competitive (Panetto *et al.*, 2016). Therefore, BPM must incorporate knowledge-intensive processes to support multi-criteria decisions by knowledge workers rather than hinder or even prevent them from working optimally by being forced into rigid processes. Additionally, effective BPM in the KE requires the use of different methodologies depending on the nature of BP (Sarnikar and Deokar, 2017; Marjanovic and Freeze, 2011). Mistakes in the assessment of the nature of BP and, consequently, the wrong choice of BPM methodologies (e.g. an attempt to strictly automate diagnostic and therapeutic processes or placing highly qualified employees on the production line without making an attempt to account for their creativity and flexibility) is one of the most serious threats to the management of value-creating BP. Therefore, as one of the first steps in preparation for the implementation or improvement of BP, it is necessary to assess the nature of BPs in order to select the appropriate method and appropriate systems to support BPM.

Carrying out such an assessment requires designing the assessment process, but, above all, defining what criteria will be taken into account and what factors affect the total assessment of each of the criteria in question. Undoubtedly, in the KE, one of the key criteria for assessing the nature of BP is the intensity of using the knowledge necessary to implement the process. However, there are no explicit and generally acknowledged criteria and measures for characterizing and distinguishing kiBP as the both academics and practitioners lack an overall common understanding of what exactly kiBP in modern organizations are. In order to effectively manage knowledge-intensive processes, we need answer the following questions: (1) How do we define BP knowledge intensity and kiBPs? (2) What criteria and measures can we use to assess the knowledge-intensity of BPs and distinguish them from other processes in the organization?

Based on the results of the qualitative systematic literature review (SLR), the aim of the article is to propose the integrated definition of kiBPs, propose a model of BP knowledge intensity and develop holistic criteria and measures for the assessment of the BPs knowledge intensity.

The next section of the paper describes the research method and the course of the conducted SLR. The third section presents the results of the review in 3 stages: levels of knowledge intensity, definitions of kiBPs and criteria for the assessment of BP knowledge intensity. In the last section, the authors will presents conclusions, limitations and suggest directions of future research.

2. Methods

The aim of the paper is to propose an integrated definition of kiBPs and a model of kiBPs together with a holistic criterion for the assessment of kiBPs. Thus, in order to achieve this complex aim, we need to establish the current state of art in this field and on this foundation build specific solutions for and conduct research on the management of kiBPs.

The objectives of the SLR are:

- (1) To identify existing definitions of the knowledge intensity and the kiBPs and develop the integrated definition of a kiBP.
- (2) To identify and select existing criteria and measures allowing for assessment of the BP knowledge-intensity and distinguishing kiBP from other BPs.
- (3) Based on the SLR results to develop a model of a kiBP together with a holistic criterion for the assessment of kiBPs.

The authors applied the qualitative SLR, as this type of literature review aims to search for, identify, select, appraise and abstract data from empirical studies to answer the research questions. They employed the typical systematic review process, but contrary to the meta-analytic approach, they used narrative and more subjective methods to bring together the findings of the included studies and follow up through research questions ([Higgins et al., 2019](#)).

Literature reviews have two main functions: (1) they serve as a background for empirical research and (2) they constitute a stand-alone piece ([Templier and Paré, 2015](#)). From the perspective of the purpose of the literature review, we divide literature reviews into four categories: description, test, extension and critique ([Paré et al., 2015](#); [Templier and Paré, 2015](#)). The aim of this article determines the choice of a stand-alone, descriptive review that tries to make sense of the existing literature by aggregating, interpreting and integrating existing research.

According to [Rousseau et al. \(2008\)](#), an SLR should have the following characteristics:

- (1) be comprehensive in the sense that it should include all relevant studies;
- (2) use transparent analyses; and
- (3) apply specific criteria to generate value from a body of previous literature.

This view is supported by [Tranfield et al. \(2003\)](#) who suggest the use of a replicable, scientific and transparent process to monitor the decisions, procedures and conclusions of the researcher and minimize bias. This approach is a way of raising the confidence of users of such pre-filtered evidence, both practitioners and academics, regarding the status of present knowledge on a given question.

The SLR process may have a varied number of phases, as the review process must be designed specifically to address the question the review is to answer ([Briner and Denyer, 2012](#)). Thus, this study has adopted the 3 generic stages of the SLR process:

- (1) Planning and scoping the search, with precise definitions of the aim and objectives to be addressed;

- (2) Rigorous execution – the identification and selection of publications and the assessment of the quality and relevance of the results;
- (3) Synthesis, analysis and reporting of the results of the review.

The above listed stages are presented below and the illustration of the whole SLR process is presented on [Figure 1](#).

2.1 Planning and scoping the search

According to [Rousseau *et al.* \(2008\)](#), a SLR should include all relevant studies and apply specific criteria to generate value from a body of previous literature. Therefore, the below presented search strategy and eligibility criteria were applied.

2.1.1 Search strategy. A systematic literature search was conducted using the ProQuest, Springer Nature and ScienceDirect full-text electronic databases. Additionally, authors scanned the reference lists of selected articles (backtracking). The authors performed an initial search (Step 1) based on the preliminary list of search keywords: “knowledge intensity,” “knowledge intensive process,” and “knowledge intensive organisation.” This initial list of the keywords resulted in a high number of publications with varying relevance to this review. To accommodate this, the authors decided to narrow the scope of the search (Step 2) to the list of the following keywords: “knowledge intensity AND measure,” “knowledge intensity AND assessment,” “knowledge intensity AND metrics”, “knowledge-intensive process AND measure”, “knowledge intensive-process AND assessment,” “knowledge-intensive process AND metrics,” “knowledge-intensive organisation AND measure,” “knowledge-intensive organisation AND assessment,” “knowledge-intensive organisation AND metrics.”

2.1.2 Eligibility criteria. In this systematic review, the authors considered peer-reviewed journals and conference proceedings published in English in the last 10 years, started the search at the beginning of March 2020 and finished it at the end of June 2020.

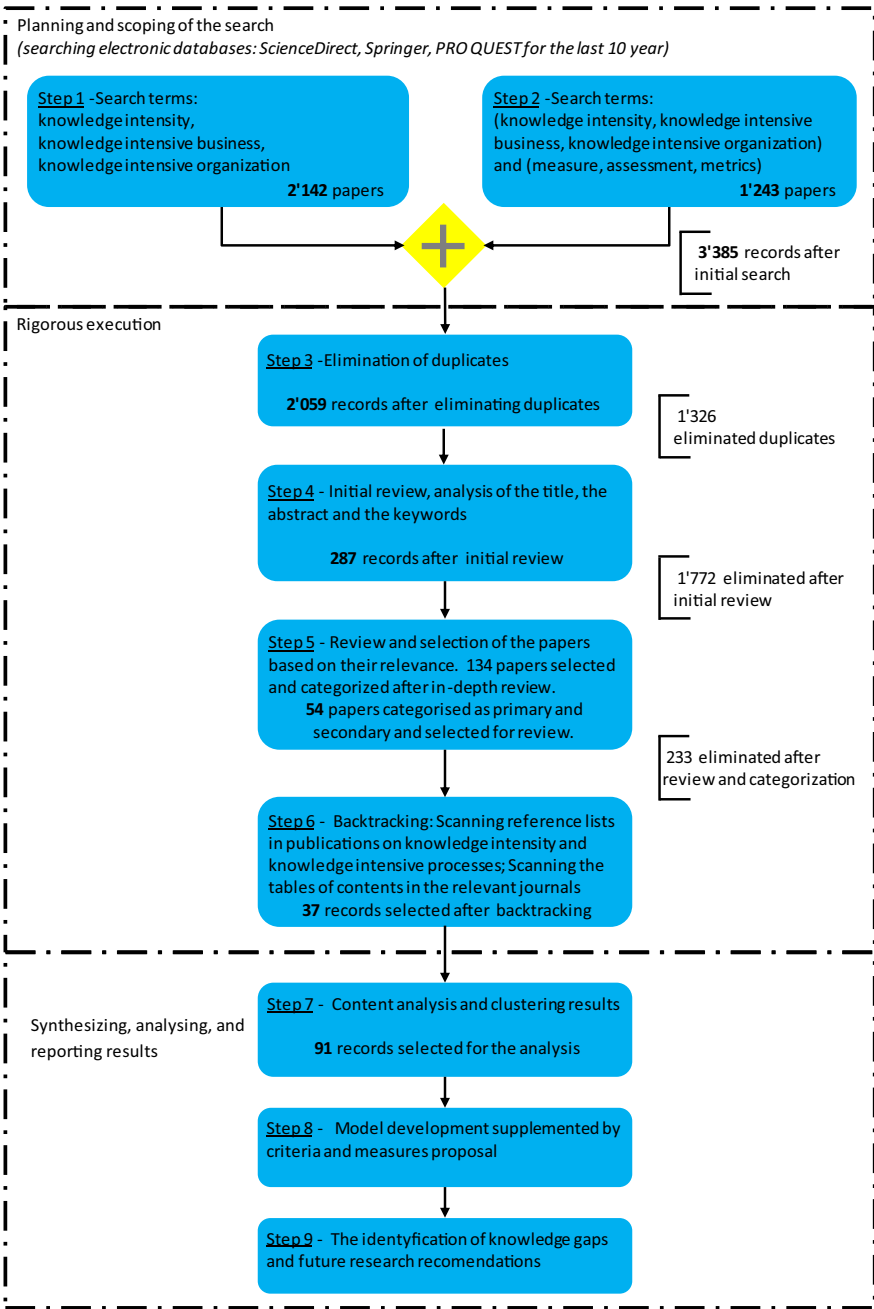
The systematic review of the literature included papers whose title and/or abstract indicated original research (theoretical and practical, as well as quantitative or qualitative or mixed method approaches) on knowledge-intensive processes, covering the definition of kiBP, as well as assessment criteria and/or measures. Studies in which knowledge-intensive processes were only a context or rationale for the conducted research, and not an end in itself, were excluded. Studies like reports, commentaries and letters were also excluded. [Table 1](#) lists inclusion and exclusion criteria for the studies.

2.2 Rigorous execution

2.2.1 Data search. The initial review covered a total of 3,385 articles published in peer-reviewed scholarly journals and conference papers. The titles and abstracts extracted from the database search were analyzed by two authors to select papers for full-text review. The authors started with the elimination of duplicates (Step 3). Then, the authors conducted the selection process (Step 4) and analyzed the title, the abstract and the keywords and searched for both:

- (1) Knowledge intensity/knowledge-intensive process/knowledge-intensive organization definition; and
- (2) Knowledge intensity measures/metrics/assessment on the organizational or process level.

The authors summarized the results of the first stage of the review and the differences were resolved by consensus. As the result of Step 4, 287 journal papers and conference papers were



Source(s): Authors' own elaboration

Figure 1.
Flow diagram of the
literature search and
study coding

selected. Thereafter (Step 5), the full texts of potentially relevant studies were reviewed to determine eligibility for inclusion. In effect, the authors selected 134 papers and categorized them based on their relevance as primary (primary research papers – on an empirical research study conducted by the authors), secondary (secondary research papers – narrative reviews, systematic reviews or a meta-analysis) and peripheral (papers which merely duplicated the content of the primary and secondary research papers or the results of which did not bring anything new). In the Results section, the authors focused on the primary and secondary categories (54 papers). Additionally, in Step 6, following the recommendations from [Greenhalgh et al. \(2004\)](#), the authors applied the citation tracking method by conducting the backtracking of knowledge intensity references from the primary and secondary category papers to make sure to cover the entire scope of field knowledge. As a result, the authors identified an additional 37 article references that were analyzed. Finally, 91 papers were selected for the in-depth review.

2.2.2 Data extraction. In the next step (Step 7), the authors analyzed all of the selected papers. Qualitative systematic reviews adopt a textual approach in the process of analysis. The reviewers may choose from a palette of analysis methods such as groupings, clusters, frameworks, classification schemes and tabulations of characteristics to synthesize the findings from the selected studies, narratively integrate the summary evidence and develop conclusions and/or recommendations ([Pare et al., 2015](#)). This review is based on the cluster and framework methods. The data extraction in form of a table was used to summarize study results. The two authors extracted the data relating to author, year, definition, criteria and measures of kiBP into a pre-formatted data collection form. After data extraction, the literature meeting the inclusion criteria was discussed and synthesized into clusters. Then, a summary of the test results was presented in the form of a model supplemented by 12 criteria with proposed measures (Step 8). Finally, the knowledge gaps were identified and future research recommendations were agreed (Step 9).

Meta-analysis was not considered appropriate for this literature review as the research question is better answered with qualitative data and opts for a narrative synthesis.

2.3 Synthesizing, analyzing and reporting the results of the review

Due to the different nature of the research results (theoretical and practical) included in this systematic review, the authors decided that the best method of synthesizing research results is narrative synthesis. In order to analyze the selected papers (Step 7), the authors divided papers into 3 clusters: (1) knowledge intensity levels, (2) definitions of kiBPs and (3) process knowledge intensity criteria and measures. Next (Step 8), each author analyzed each of the 3 clusters and then each author reviewed the synthesis of the analyzed articles of the other author and provided a critical evaluation. Afterward, summarizing of included studies in a narrative synthesis within a model and framework was performed. Finally (Step 9), the

Criteria	Inclusion	Exclusion
Time period	After 2010	Before 2010
Language	English	Other languages
Type of source	Peer-reviewed journals and conference proceedings	Non-peer reviewed journals and conference proceedings
Type of studies	Original studies	Reports, commentaries, letters
Aim: To identify the definition of kiBP and assessment criteria and/or measures	Literature covers the definition of kiBP and assessment criteria and/or measures	Literature does not cover the definition of kiBP and assessment criteria and/or measures
Source(s): Authors' own elaboration		

Table 1.
Inclusion and
exclusion criteria for
the studies

authors identified knowledge gaps and recommended future research directions. Results of the review will be presented in the next section.

3. Results

The analysis of the papers selected for the in-depth review focused on developing the classification of knowledge intensity levels, defining knowledge-intensive processes and proposing criteria and measures for the assessment of BPs knowledge intensity. The results are presented in the following subsections.

3.1 Knowledge intensity levels

According to the [Cambridge Dictionary \(n.d.\)](#), knowledge-intensive means “*needing a lot of experience, understanding, information, and skills in order to be successful*”. Knowledge intensity can also be defined as an expansion of the utilization of knowledge potential ([Očenášková et al., 2012](#)) and it can be provided not only at the organizational level, but also in the case of all sectors, including national economics ([Chen and Dahlman, 2005](#); [Očenášková et al., 2012](#)). Knowledge intensity can be considered as a distinctive characteristic of:

- (1) *The economy or country* – [Narula \(2015\)](#), [Low et al. \(2015\)](#), [Long et al. \(2017\)](#), [Grigorescu et al. \(2020\)](#);
- (2) *Particular industry* – [Villar et al. \(2012\)](#), [Nandan and Ciccotosto \(2014\)](#), [Collet et al. \(2015\)](#), [Dong et al. \(2019\)](#);
- (3) *Services* – [Rosenbaum \(2013\)](#), [Edgar et al. \(2014\)](#), [Ryzhkova \(2015\)](#), [Heikka and Natti \(2018\)](#).
- (4) *Organization or company* – [Makani and Marche \(2010\)](#), [Seto et al. \(2011\)](#), [Kapyła et al. \(2011\)](#), [Carmel et al. \(2013\)](#), [Bolisani and Scarso \(2015\)](#), [van Greunen et al. \(2019\)](#), [Mazorodze and Buckley \(2019\)](#);
- (5) *Processes* – [Vaculin et al. \(2011\)](#), [Isik et al. \(2012\)](#), [Di Ciccio et al. \(2015\)](#), [Little and Deokar \(2016\)](#), [Santoro and Baiao \(2018\)](#);
- (6) *Tasks or activities* – [Sarnikar and Deokar \(2017\)](#), [Vaculin et al. \(2011\)](#), [Shen et al. \(2012\)](#), [Lin et al. \(2015\)](#), [Musto et al. \(2015\)](#), [Baumeister et al. \(2016\)](#);
- (7) *or even an individual worker* – [Massaro \(2012\)](#), [Sheikh and Lim \(2015\)](#), [Kianto et al. \(2019\)](#).

3.2 The definition of knowledge-intensive business processes

From the BPs perspective, knowledge intensity can be defined as an extent to which knowledge processes are performed and knowledge resources are utilized ([Očenášková et al. 2012](#)). However, as highlighted by [Unger et al. \(2015\)](#), to date, there exists no commonly agreed definition of kiBPs. Even the spelling or abbreviations relating to this concept are very diversified (e.g. KnowiP, kiBP, KIBP or KIPs). The review of the existing definitions is presented in [Table 2](#).

The chronological review of kiBP definitions reveals that they are becoming more and more extensive over time and, apart from the very nature of the processes, they take into account the role of knowledge, the role of the knowledge worker and the environment in which the process is planned and executed. In general, the definitions of kiBPs focus on three main overlapping domains.

- (1) *The role of knowledge* (in achieving a goal or creating value through a process). Most often this domain pays attention to value creation only through the use of mostly tacit

No	Autor	Year	kiBP definition
1	Davenport, Jarvenpaa and Beers	1996	... the diversity and uncertainty of process input and output
2	Eppler, Seifried and Ropnack	1999	... business processes that have a high need for innovation of single agent who affects the results of a process directly
3	Heisig	2002	... those processes that cannot be planned in advance or can only be planned to a limited extent with regard to the knowledge they require
4	Remus	2002	... processes characterized by a number of features that do not or only partially allow for an improvement according to classic methods of business process optimization (BPO) or more radical methods of business process reengineering (BPR)
5	Papavassiliou and Mentzas	2003	The processes considered are often complex in general, with many, but conceptually simple, (usually) document-centered activities; at the heart of these processes are few central decision steps which require personal judgment based on experience, a comprehensive knowledge about the given as well as about older, similar cases, access to much specific information in files and forms, manifold legal regulations and standard operating procedures etc
6	Gronau and Weber	2004	... its value can only be created through the fulfillment of the knowledge requirements of the process participants. Clues for a knowledge-intensive process are apart from the above mentioned criteria – Diversity of information sources and media types – Variance and dynamic development of process organization – Many process participants with different expert's reports – Use of creativity – High degree of innovation – An available degree of decision scope
7	Richter-von Hagen, Ratz and Povalej	2005	... sequences of activities based on knowledge-intensive acquisition and handling
8	Bahrs and Müller	2005	... task sequences, which strongly rely on the employment of tacit knowledge
9	Jochem, Geers and Heinze	2011	... processes characterized by their high amount of exceptions. Knowledge bound to these processes' participants is the basic ingredient for the value creation. Activities within these processes cannot be automated or standardized
10	Vaculin <i>et al.</i>	2011	... processes whose conduct and execution are heavily dependent on knowledge workers performing various interconnected knowledge-intensive decision-making tasks
11	Di Ciccio, Marrella and Russo	2012	... process characterized by activities that cannot be planned easily, may change on the fly and are driven by the contextual scenario that the process is embedded in
12	Isik, van der Bergh and Martens	2012	... processes that require very specific process knowledge, typically expert involvement, that are hard to predict and vary in almost every instance of the process
13	Unger, Leopold and Mendling	2015	... processes characterized by high variability, unforeseeable exceptions, the need for creativity and complex tasks
14	Rodrigues <i>et al.</i>	2015	... process characterized as a bag of activities based on knowledge-intensive acquisition and manipulation. A KIP differentiates from traditional business processes mainly because of the dynamism or the variability of event flows among the process instances

(continued)

Table 2.
The review of
definitions of
knowledge-intensive
business processes

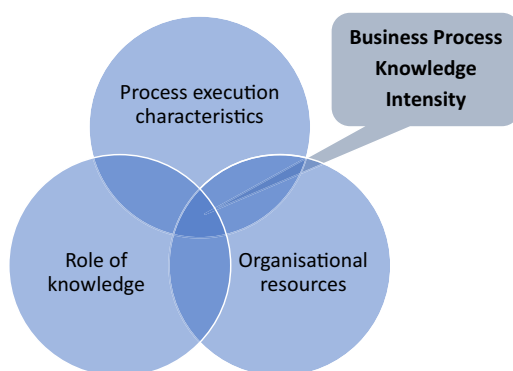
No	Autor	Year	kiBP definition
15	Di Ciccio, Marrella and Russo	2015	... processes whose conduct and execution are heavily dependent on knowledge workers performing various interconnected knowledge-intensive decision-making tasks
16	dos Santos Franca <i>et al.</i>	2015	.. typically comprises sequences of activities based on intensive acquisition, sharing, storage and (re)use of knowledge, so that the amount of value added to the organization depends on the knowledge of the process executors. They are naturally more complex, since they deal with diffuse and tacit definitions, unpredictable decisions, creativity-oriented tasks and paths, dynamic execution that evolves based on the experience acquired by the executors
17	Little and Deokar	2016	... a process that requires an individual's judgment based on that individual's experiences and knowledge obtained through a variety of sources such as knowledge repositories or experts. KIBP can be viewed as a collection of related and often interdependent activities that cannot be fully predetermined, as they often entail innovation on the part of the individual, involve further complex tasks
18	Santoro and Baiao	2018	... process characterized by the presence of collaborative interactions among participants and flexibility to perform work, making the process less predictable than a structured routine

Table 2. Source(s): Authors' own elaboration

- knowledge, the creation of new knowledge during the implementation of the process, as well as the creation of innovations or adaptations resulting from the application of knowledge directly affecting the result/outcome of the process (e.g. [Richter-von *et al.* \(2005\)](#), [Bahrs and Müller \(2005\)](#), [Jochem *et al.* \(2011\)](#), [Vaculin *et al.* \(2011\)](#), [Isik *et al.* \(2012\)](#), [dos Santos Franca *et al.* \(2015\)](#), [Little and Deokar \(2016\)](#)).
- (2) *The process execution characteristics*. The most common ones are: unpredictability, flexibility, uniqueness, contextuality, complexity, a large number of exceptions, or a course dependent on decisions based on knowledge of the process executors (e.g. [Davenport *et al.* \(1996\)](#), [Eppler *et al.* \(1999\)](#), [Heisig \(2002\)](#), [Papavassiliou and Mentzas \(2003\)](#), [Remus \(2002\)](#), [Gronau and Weber \(2004\)](#), [Jochem *et al.* \(2011\)](#), [Di Ciccio *et al.* \(2012\)](#), [Isik *et al.* \(2012\)](#), [Unger *et al.* \(2015\)](#), [Rodrigues *et al.* \(2015\)](#), [dos Santos Franca *et al.* \(2015\)](#), [Little and Deokar \(2016\)](#), [Santoro and Baiao \(2018\)](#)).
- (3) *Organizational resources* (needed for processes planning and execution). This domain also refers to information and data reservoirs and communication channels, but most it refers to human and social capital – employees who perform processes and their relations. It emphasizes, above all, the requirement of creativity and commitment, cooperation skills, as well as comprehensive expert knowledge (e.g. [Papavassiliou and Mentzas \(2003\)](#), [Gronau and Weber \(2004\)](#), [Vaculin *et al.* \(2011\)](#), [Unger *et al.* \(2015\)](#), [Little and Deokar \(2016\)](#), [Santoro and Baiao \(2018\)](#)).

The presented results of the SLR indicate that kiBP are processes in which the delivered value depends on the creative and innovative use of knowledge by process executors in accordance with the context of execution and based on the organizational resources.

Starting from the above three domains, we can define the BPs knowledge intensity as a set of characteristics encompassing process execution, the role of knowledge and organizational resources. This set of three groups of characteristics creates a foundation to frame or characterize the nature of BP from the knowledge intensity perspective ([Figure 2](#)).



Source(s): Authors' own elaboration

Figure 2.
The business process
knowledge
intensity model

The most important element of kiBP are "... *interconnected knowledge intensive decision-making tasks*" (Vaculin *et al.*, 2011). Their execution requires "... *intensive acquisition, sharing, storage and (re)use of knowledge, so that the amount of value added to the organization depends on the knowledge of the process executors*" (dos Santos Franca *et al.*, 2015). Sarnikar and Deokar (2017, p. 701) define a knowledge-intensive task "*as requiring high agent innovation, involving multiple decision paths, contingent upon numerous eventualities and being highly dependent on agent actions. They are also characterized by long learning time to perform the task and lower knowledge half-life, where knowledge quickly becomes obsolete.*" In other words, those tasks require significant user knowledge and creativity for their effective enactment (Marjanovic and Seethamraju, 2008; Isik *et al.*, 2013). Such tasks are determined at run-time instead of strictly modeled at design time and have to be executed depending on the actual context of the process instance, allowing for resource allocation, including knowledge allocation and supporting the user in decision-making, while checking for integrity based on constraints and guidelines to ensure consistency and compliance (Feldkamp *et al.*, 2007). Consequently, they have a direct impact on the course of processes and the value they provide. Also, the future competitive position of the organization depends on the experience and new knowledge accumulated during their implementation by process executors.

3.3 Criteria used to assess the business processes knowledge intensity

Starting from the BP knowledge intensity model, the next step was the identification of criteria determining the BP knowledge intensity level. Identified on the basis of the literature review 12 criteria were clustered into 3 domains. These criteria will be presented and defined below.

3.3.1 Criteria for assessing the role of knowledge in a kiBP. According to the reviewed literature, kiBPs are those in which: tacit knowledge is more significant with respect to process execution than codified knowledge and is prime driver for value creation; process execution requires the verification of old knowledge and the creation of new knowledge to meet the high demand for innovation and creativity; extensive learning time is required to acquire knowledge for executing the process; knowledge needed for process execution is very diversified and replaced faster than the process execution time frame. Based on the above, four criteria of BP knowledge intensity were identified: the role of tacit knowledge, the role of new knowledge, the process knowledge acquisition time and the knowledge scope and replacement rate. The criteria, their definitions, measures and sources from which they were formulated are presented in Table 3.

Criteria	Definition	Measures	Source
The role of tacit knowledge	Tacit knowledge, including experience, is more significant with respect to process execution than codified knowledge and is the prime driver for value creation	Percentage of the total number of process executions in which tacit knowledge is used Weight of tacit knowledge usage in achieving goals and delivering value through the process	Windrum and Tomlinson (1999), Autio <i>et al.</i> (2000), Maracine and Scarlet (2009), Muller and Doloreux (2009), Isik <i>et al.</i> (2012)
The role of new knowledge	Process execution requires the verification of old knowledge and the creation of new knowledge, as there is high demand for innovation and creativity (organizations produce and sell knowledge)	Percentage of the total number of process executions in which new knowledge is created and used Weight of new knowledge usage in achieving goals and delivering value through the process	Autio <i>et al.</i> (2000), Gronau and Weber (2004), Horgos and Koch (2008), Marciane and Scarlet (2009), Kapyla <i>et al.</i> (2011)
The Process knowledge acquisition time	Extensive learning time is required to acquire knowledge for executing the process	Time and costs of formal, organizational transfer of knowledge (internships, trainings, meetings) and knowledge acquisition The time spent by process executors to update their essential knowledge (e.g. analyze the available information on experiences from process execution from the intranet and social media, as well as familiarize themselves with case studies etc.)	Eppler <i>et al.</i> (1999), Bhatt and Grover (2005)
The knowledge scope and replacement rate	Knowledge needed for process execution is very diversified and replaced faster than the process execution time frame	The number of knowledge areas the use of which is necessary to achieve the goals of the process The average time between the need to update the systems supporting the implementation of processes (e.g. workflow, RPA or AI) The average cost of the necessary updates of systems supporting the execution of processes	Eppler <i>et al.</i> (1999), Jung <i>et al.</i> (2007), Isik <i>et al.</i> (2013)

Table 3.
Criteria of the role of knowledge for assessment of the kiBP **Source(s):** Authors' own elaboration

3.3.2 *Criteria for assessing process execution characteristics in a kiBP.* Based on the reviewed literature, the authors concluded that kiBPs are those in which: process execution is tied to the broad scope of necessary and possible decisions, which are often complex in nature; the process is highly unpredictable due to the context of execution or the required process input and output; process design and execution requires a high degree of creativity and innovativeness; the complexity and the multidisciplinary nature of the process requires an extensive integration of different knowledge domains. As the result of the above,

characteristics of the four criteria of BP knowledge intensity were identified: the scope of decisions, predictability and repeatability, level of creativity and innovation and process complexity and contextuality. This set of criteria, their definitions, measures and sources from which they were formulated are presented in [Table 4](#).

3.3.3 Criteria for assessing organizational resources in a kiBP. The authors of the reviewed papers point out that kiBPs are those in which: process design and execution are heavily dependent on knowledge workers; the process environment creates organizational capabilities to transform and create knowledge; in the whole BPM lifecycle there is a high diversity of information sources and intensity of internal network communication; process design and execution require broad and high quality professional relations with a range of stakeholders. As the result of the above characteristics, the last set of criteria of BP knowledge intensity was identified: the role of knowledge worker, organizational resources, communication diversity and intensity in the internal network and social capital. These criteria, their definitions, measures and the sources from which they were formulated are presented in [Table 5](#).

The set of 12 criteria presented in this section allows the holistic assessment of BP knowledge intensity. Detailed measures of the individual criteria of the kiBP assessment significantly depend on the contextual factors of the process implementation, such as the industry or process maturity of the organization, and even the systems supporting BPM may concern the effectiveness of the implemented BPM and improvement perspective ([Kapyła et al., 2011](#); [Sidorova and Isik, 2010](#); [Lonngvist, 2004](#); [Autio et al., 2000](#)). For most criteria, the following measures can be formulated:

- (1) Frequency of occurrence (e.g. as the number of occurrences in a defined unit of time or as a percentage of the total number of process executions meeting a given criterion),
- (2) Weights of the conditions specified in the criterion to achieve the goals and deliver value through the process (e.g. using seven-point Likert scale),
- (3) Time and costs of codification and sharing in the organization the knowledge revealed or created during the process, including information about the knowledge negatively verified during the process,
- (4) Time and costs of updating systems supporting the implementation of processes (e.g. workflow, RPA or iBPMS with ML/AI elements) in order to take into account the knowledge disclosed or created during the process.

4. Discussion and conclusions

Over the last few decades, knowledge has been recognized as one of the key competitiveness factors of business organizations. In the KE, organizations implement more and more knowledge-intensive processes like customer support, design of new products/services, marketing, data quality management, IT governance and strategic management ([Marjanovic and Freeze, 2011](#)). Thus, organizations are becoming increasingly conscious of the necessity to formalize and capitalize the knowledge produced and mobilized by their BP in order to improve their competitive advantage ([Ben Hassen et al., 2016](#)). Consequently, an increase in the interest in managing kiBP through the use of knowledge management systems and the integration of these systems with BPM can be observed ([Marjanovic and Freeze, 2011](#); [Sarnikar and Deokar, 2017](#)). As suggested by [Brucker et al. \(2016\)](#), kiBP can be perceived as a solution for all organizations in which processes are no longer structured, longstanding and repeatable, which continuously evolve and have to deal with missing, uncertain and erroneous data. However, research on kiBP, although steadily increasing, is still limited and focused on narrow aspects of these phenomena. Consequently, there are no explicit and

Criteria	Definition	Measures	Sources
The scope of decisions	Process execution is tied to the broad scope of necessary and possible decisions, which are often complex in nature	Number of decisions in the process made by process executors (human)	Eppler <i>et al.</i> (1999), Jones (2001), Papavassiliou and Mentzas (2003), Abecker (2004), Lai and zur Muehlen (2004), Gronau and Weber (2004), Marjanovic and Seethamraju (2008), Reichert and Weber (2012), Di Ciccio <i>et al.</i> (2015), Venero <i>et al.</i> (2019)
	Such processes require a significant amount of freedom to operate and apply knowledge in order to adapt to various contexts	Number of decisions in the process in which unpredictable, ad hoc exceptions may occur	
Predictability and repeatability	The process is highly unpredictable (event-driven) due to the context of execution or the required process input and output	Percentage of structured processes at the N ($N-1$) level Number of events within the process that may occur unpredictable or ad hoc or whose occurrence may require unpredictable decisions before the process is carried out Weight of unpredictable events for achieving goals and delivering value through the process	Abecker (2004), Gronau and Weber (2004), Ditillo (2004), Richter-von Hagen <i>et al.</i> (2005), Marjanovic and Seethamraju (2008), Marjanovic and Freeze (2011), Panian (2011), Di Ciccio <i>et al.</i> (2012), Unger <i>et al.</i> (2015)
Level of creativity and innovation	Process design and execution requires a high degree of creativity and innovativeness	Number of reported innovations in a defined time unit Weight of identified innovations for achieving goals and delivering value through the process	Eppler <i>et al.</i> (1999), Strambach (2001), Gronau and Weber (2004), Richter-von Hagen <i>et al.</i> (2005), Harmon (2007), Marjanovic and Seethamraju (2008), Sarnikar and Deokar (2017), Isik <i>et al.</i> (2013), Di Ciccio <i>et al.</i> (2012), Unger <i>et al.</i> (2015)
Process complexity and contextuality	The complexity and the multidisciplinary nature of the process requires an extensive integration of different knowledge domains. The context of process execution may have a significant impact on the execution of the process and its results	Number of knowledge areas the use of which is necessary to achieve the goals of the process Number of stakeholders involved in the implementation of the process	Eppler <i>et al.</i> (1999), Harmon (2007), Swart and Kinnie (2003), Abecker (2004), Gronau and Weber (2004), Bhatt and Grover (2005), Marjanovic and Seethamraju (2008), Davenport (2015), Marjanovic and Freeze (2011), Panian (2011), Isik <i>et al.</i> (2013), Unger <i>et al.</i> (2015), Little and Deokar (2016), Venero <i>et al.</i> (2019)

Table 4.
Criteria of process execution for assessment of the kiBP **Source(s):** Authors' own elaboration

generally acknowledged kiBP definition and criteria with measures for differentiating kiBP from other processes, and therefore, the concept of knowledge-intensity has no significant managerial use (Kapyla *et al.*, 2011). Thus, the aim of the article was to propose the integrated

Criteria	Definition	Measures	Source
The role of the knowledge worker	Process design and execution are heavily dependent on knowledge workers. The process executors are held highly accountable for decisions, use novel knowledge to solve complex problems, and thus have a large influence on the flow, and, in effect, the end result of the process	Number of decisions in the process made by knowledge workers Number of decisions in the process in which unpredictable, ad hoc exceptions may occur Weight of decisions made in unpredictable and ad hoc exceptions for achieving goals and delivering value through the process	Alvesson (2001), Swart and Kinnie (2003), European Parliament (2008), Marjanovic (2010), von Nordenflycht (2010), Heidary Dahoie <i>et al.</i> (2011), Kapyla <i>et al.</i> (2011), Di Ciccio <i>et al.</i> (2015), Unger <i>et al.</i> (2015), Ramadan <i>et al.</i> (2017), Venero <i>et al.</i> (2019)
Organizational assets	The process environment (organizational structure, organization culture, education and the professionalization of the workforce, work practices, knowledge sharing, continuous learning) create organizational capabilities to transform and create knowledge in the whole BPM lifecycle	Organization's process maturity level for groups of processes or key processes BPM maturity level in the organization	Abecker (2004), Gronau and Weber (2004), Kapyla <i>et al.</i> (2011), Monavvarian <i>et al.</i> (2013), Unger <i>et al.</i> (2015), Ramadan <i>et al.</i> (2017)
Communication diversity and intensity in the internal network	There is a high diversity of information sources and intensity of internal network communication	Number of information sources required for the process execution Number of communication channels used for the process execution Number of discussion groups (e.g. in Intranet, social media, etc.)	Abecker (2004), Gronau and Weber (2004), Lonngvist (2004), Smith <i>et al.</i> (2005), Dalmaris <i>et al.</i> (2007), Kulkarni and Ipe (2007), Marjanovic <i>et al.</i> (2007), Schymik <i>et al.</i> (2007), Marjanovic and Freeze (2011), Santoro and Baiao (2018)
Social capital	Process design and execution require broad and high quality professional relations (both formal and informal) with a range of stakeholders	Number of stakeholders involved in the design and modeling of the process Number of stakeholders involved in the implementation of the process Time spent by process executors to exchange knowledge with external professionals (e.g. experts, consultants, etc.)	Swart and Kinnie (2003), Dalmaris <i>et al.</i> (2007), Marjanovic <i>et al.</i> (2007), Kapyla <i>et al.</i> (2011), Isik <i>et al.</i> (2013), Ramadan <i>et al.</i> (2017), Scaringella and Radziwon (2017)

Source(s): Authors own elaboration

Table 5.
Criteria of
organizational
resources for
assessment of the kiBP

definition of kiBPs, propose a model of BP knowledge intensity and develop holistic criteria for the assessment of the BPs knowledge intensity.

Based on the qualitative SLR of the identified papers meeting the inclusion criteria, the authors classified the existing definitions of knowledge intensity, starting from knowledge-

intensive economy or country, sectors and services, through knowledge-intensive organizations, processes and tasks to knowledge-intensive workers. As the focus of the paper is on kiBPs, an in-depth review of kiBP definitions was conducted and an integrated definition of a kiBP was proposed. The major value of this paper is the proposal of the BP knowledge intensity model consisting of three domains:

- (1) The role of knowledge in achieving a goal or creating value through a process.
- (2) The process execution (performance) characteristics; and
- (3) Organizational resources needed for processes planning and execution.

A list of criteria and measures enabling a multi-faceted assessment of the BP knowledge intensity, consistent with the model and possible to verify in practice, was also presented.

4.1 Future research

The conducted research revealed a number of gaps in the literature and in future research it would be useful to conduct SLR for the knowledge intensity of both business organizations and tasks, as well as an in-depth analysis of the impact of each identified knowledge intensity criteria on the BP knowledge intensity. It would be also valuable to conduct research on the dynamics of the structure and nature of BPs in organizations from the knowledge intensity perspective. In further works the authors intend to further develop a framework for an objective assessment of the BPs knowledge intensity, enabling the selection of methods and techniques for implementing BPM consistent with the real nature of processes or groups of processes in the organization.

4.2 Limitations

This review has certain limitations. First, the literature review was based on a set of predefined keywords and only peer-reviewed sources published in a limited time frame and written in English were considered, excluding potentially valuable theses and dissertations, non-English journals, non-academic reports and articles, and gray literature not published by commercial publishers. Second, the review employed qualitative, narrative synthesis and in the future it would be valuable to consider other approaches, such as bibliometric data analyzed in the “VOSviewer” software.

4.3 Practical and theoretical contribution

The authors believe that the study makes a significant and inspiring contribution to the theoretical reflection on the existing state of knowledge on knowledge intensity, and especially, BP knowledge intensity. The results of the paper also contribute to future research in the field by defining BP knowledge-intensity and equipping researchers and practitioners with a set of criteria and measures that allow for the holistic assessment of process knowledge intensity and thus allows for the future diagnosis of the nature of knowledge-based BPs from the knowledge intensity perspective, as well as for dynamic changes to the structure of BPs. By creating a broader model for BP knowledge intensity assessment, the authors also contribute to the more effective management of kiBP and help find similarities and differences between various kiBPs, as well as provide useful implications for their management and research.

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